Does Climate Change Reduce Global Growth and Welfare?

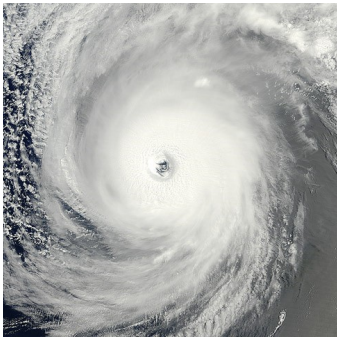
Diego R. Känzig
Northwestern University, CEPR & NBER

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Human activity is warming the planet and altering ecosystems





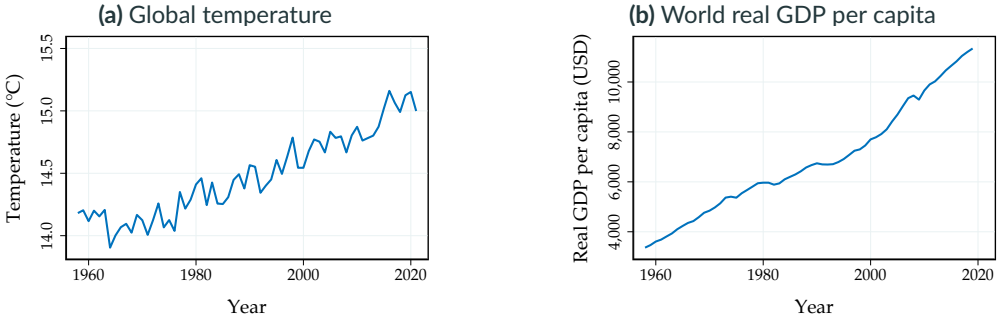


How large are the economic losses from climate change?

Evidence on economic losses remains unsettled

- Influential empirical work compares **hotter and cooler places over time**
  - Many estimates imply **moderate average GDP losses**: around 1–3% per 1°C
- But local variation can miss common shocks, spillovers, and ocean dynamics
- In macroeconomics, the key object is the effect of a **global climate shift**
- Recent work exploits **global temperature variation** to estimate climate damages
  Bilal & Känzig (QJE 2026), Bilal, Känzig & Lisiecki (2026)

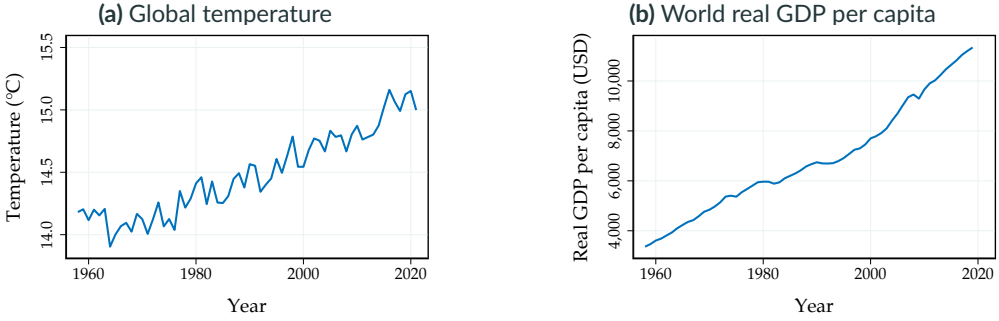
Global temperature and economic growth



Notes: Global average temperature including sea surface from Berkeley Earth; world real GDP per capita from PWT.

- The shared trend obscures the causal effect of warming
- We therefore focus on **temperature shocks**: El Niño, volcanic eruptions, solar cycles, …

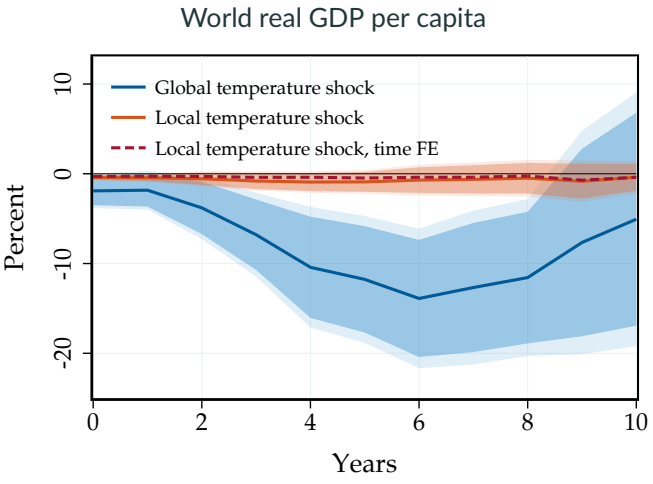
Global temperature and economic growth



Notes: Global average temperature including sea surface from Berkeley Earth; world real GDP per capita from PWT.

- The shared trend obscures the causal effect of warming
- We therefore focus on **temperature shocks**: El Niño, volcanic eruptions, solar cycles, …
- Trace effect on GDP and mortality for the following decade

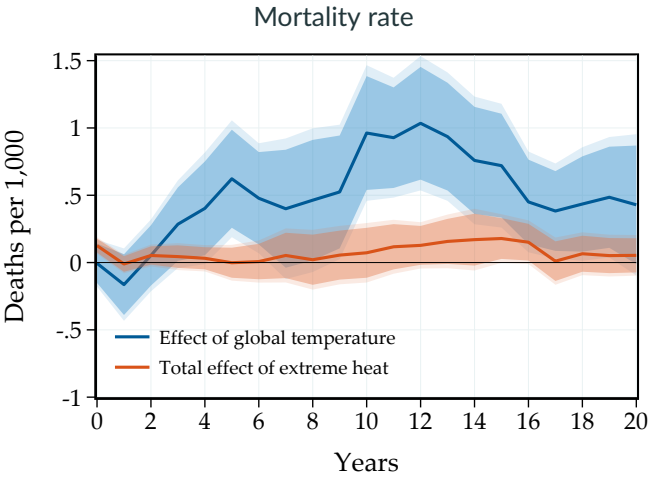
The impact of a 1°C global temperature shock



Notes: Point estimates with 90 and 95% confidence bands.

- A 1°C rise leads to fall in world GDP per capita by **over 10%**
- **Local shocks** have only modest effects on average
- **Global shocks** capture broader effects through damaging extremes and spillovers

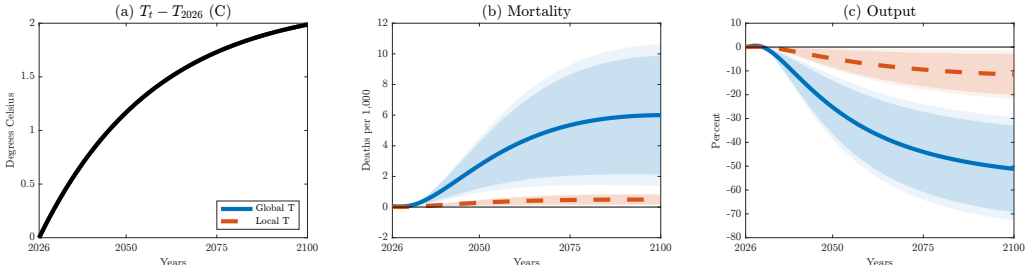
The impact of a 1°C global temperature shock on mortality



Notes: Point estimates with 90 and 95% confidence bands.

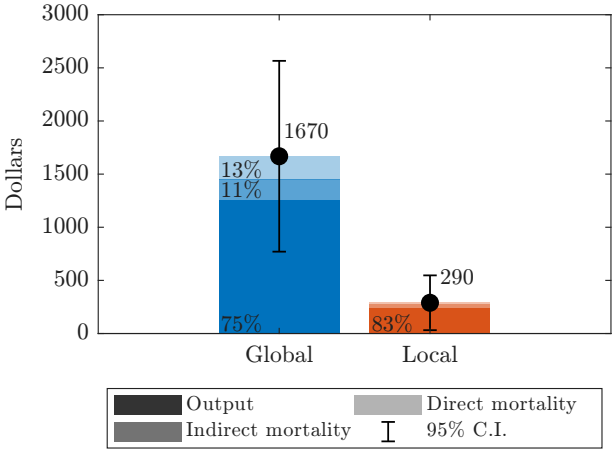
- Mortality rises by **1 death per 1,000 people per year**
- Much larger than deaths from extreme **heat exposure** alone
- Income losses also contribute to higher mortality

The impact of climate change



- About **2°C additional warming by 2100**, relative to 2026, in a standard growth model
- **Over 50% lower output** in the long run
- Mortality rises by **6 excess deaths per 1,000** people per year

The Social Cost of Carbon



- Total estimate over **$1,600/tCO$_2$**
- Economic losses account for **three quarters**; mortality for **one quarter**
- Much higher than **local** temperature estimates

Climate change reduces growth and welfare

- We document **climate damages 5-6 times larger** than previous estimates
- Our estimates capture broader impacts through **extremes** and **spillovers**
- Projected losses are comparable to a permanent 1929 Great Depression
- **Unilateral decarbonization** policy is optimal for large and populous economies

Thank you!